PROJECT SUBMISSION (CODES AND OUTPUTS)

Code Imported on Google Colab with libraries :

```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
from google.colab import files
uploaded = files.upload()
```

```
Choose Files Power BI U...Data (1).xlsx
Power BI Updated Data analyst Data (1).xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 374771 bytes, last modified: 5/8/2026 - 100% done
Saving Power BI Updated Data analyst Data (1).xlsx to Power BI Updated Data analyst Data (1).xlsx
```

```
pip install seaborn openpyxl
```

```
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-packages (3.1.5)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.0.2)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /usr/local/lib/python3.12/dist-packages (from seaborn) (3.10.0)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.0.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2026.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
```

```
# LOAD & CLEAN DATA
df = pd.read_excel('Power BI Updated Data analyst Data (1).xlsx')
# Clean column names
df.columns = df.columns.str.strip()
# Clean text columns
df['Leadership- skills'] = df['Leadership- skills'].str.strip().str.lower()
df['Family Income'] = df['Family Income'].str.strip()
# Map Family Income to numeric midpoints for correlation
```

```
income_map = {'0-2 Lakh': 1, '2-5 Lakh': 3.5, '5-7 Lakh': 6, '7 Lakh+': 8}
df['Family Income Numeric'] = df['Family Income'].map(income_map)
# Colour palette
COLORS = ['#4361ee', '#f72585', '#4cc9f0', '#7209b7', '#3a0ca3',
          '#f3722c', '#90be6d', '#43aa8b', '#577590', '#f8961e']
sns.set_style("whitegrid")
print("=" * 60)
print("  CLOUD COUNSELAGE GPI — DATA ANALYSIS SOLUTIONS")
print("=" * 60)
```

(A) BASIC QUESTIONS (Attempt any 8):

1. How many unique students are included in the dataset?

CODE :

unique_students = df['Email ID'].nunique()

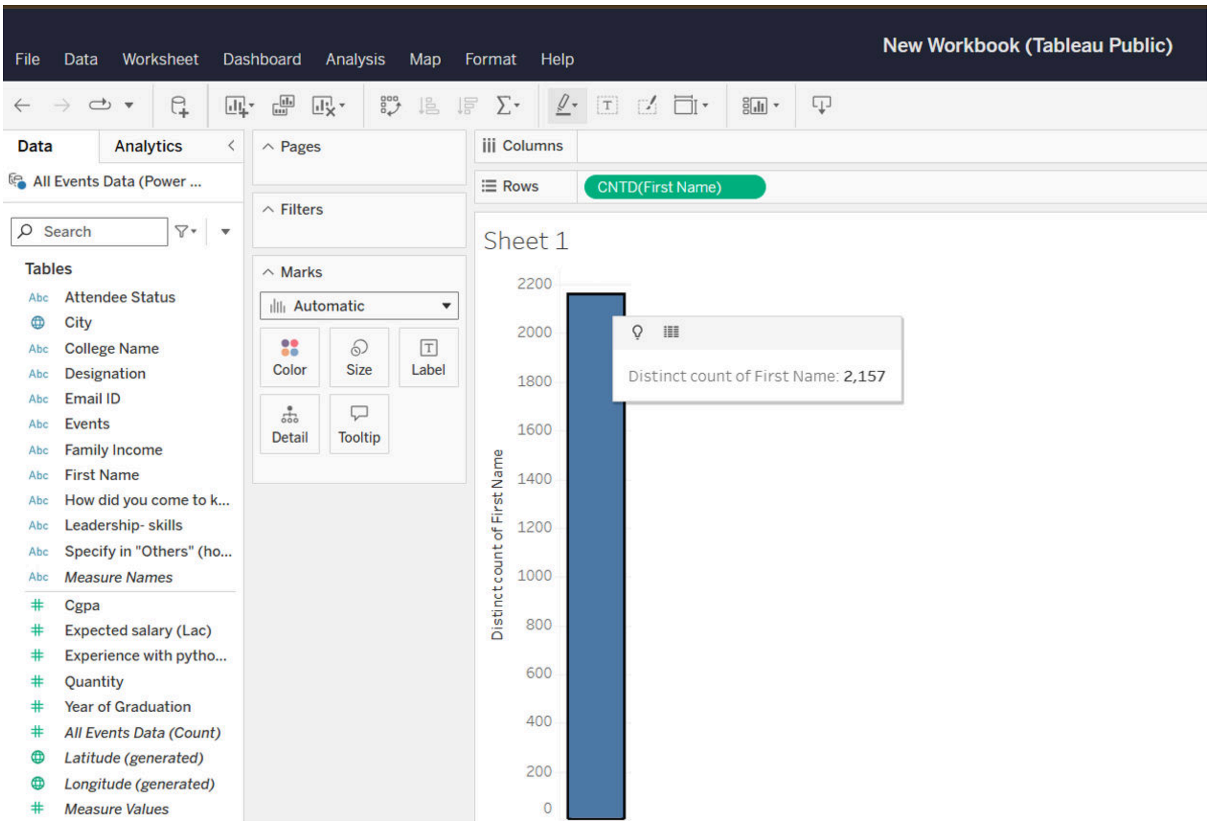
print(f"\n Unique students in dataset: {unique_students}")

CONCLUSION : (Colab Output)

```
============================================================
  CLOUD COUNSELAGE GPI — DATA ANALYSIS SOLUTIONS
============================================================
```

```
Unique students in dataset: 2157
```

CONCLUSION : (Tableau Output)



2. What is the average GPA of the students?

CODE :

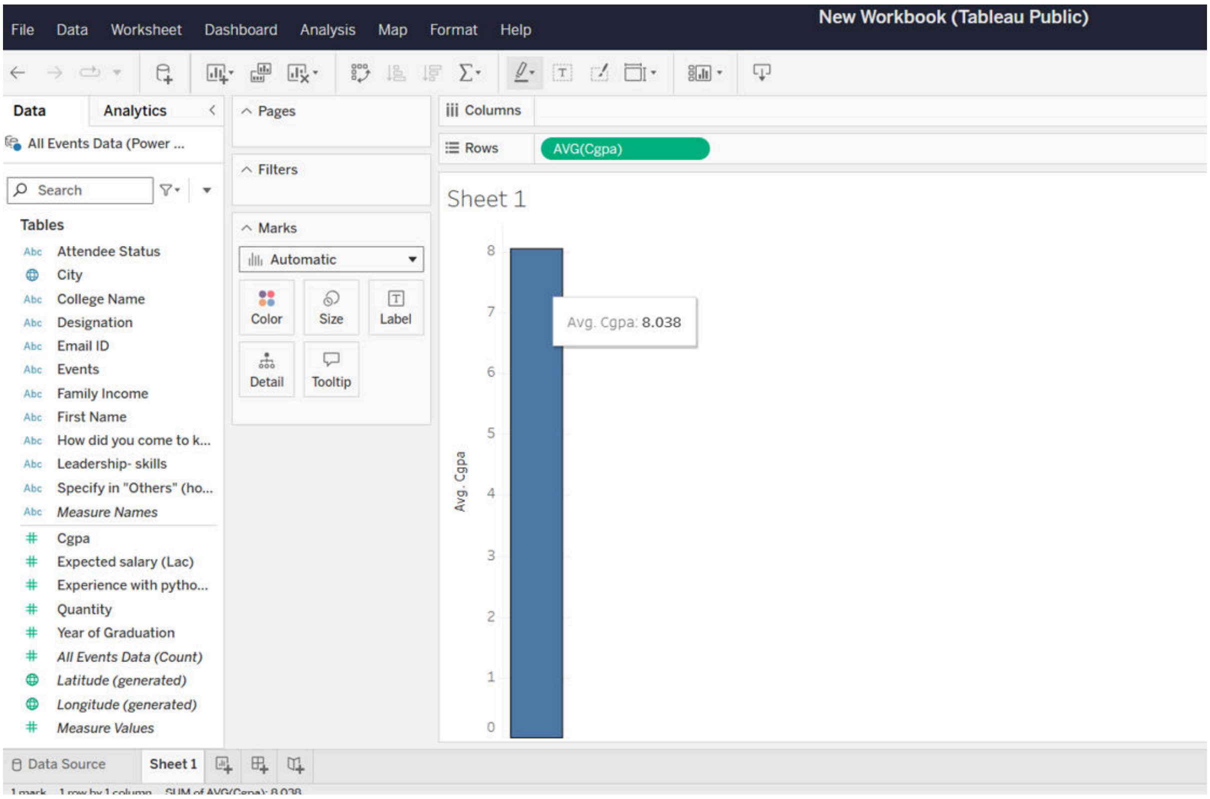
avg_gpa = df['CGPA'].mean()

print(f"\n Average GPA of students: {avg_gpa:.2f}")

CONCLUSION :(Colab Output)

```
Average GPA of students: 8.04
```

Tableau Output :



3. What is the distribution of students across different graduation years?

CODE :

```
grad_dist = df['Year of Graduation'].value_counts().sort_index()

print(f"\n Distribution by Graduation Year:\n{grad_dist.to_string()}")

fig, ax = plt.subplots(figsize=(8, 5))

bars = ax.bar(grad_dist.index.astype(str), grad_dist.values, color=COLORS[:len(grad_dist)],
edgecolor='white', linewidth=0.8)

for bar in bars:

    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 10,

            str(int(bar.get_height())), ha='center', va='bottom', fontsize=10, fontweight='bold')

ax.set_title("Q3 — Students by Graduation Year", fontsize=14, fontweight='bold', pad=15)

ax.set_xlabel("Graduation Year"); ax.set_ylabel("Number of Students")

plt.tight_layout()

plt.show()
```

CONCLUSION : (Colab Output)

```
Distribution by Graduation Year:
```

```
 Year of Graduation
 2023     1536
 2024     1511
 2025     1292
 2026      555
```

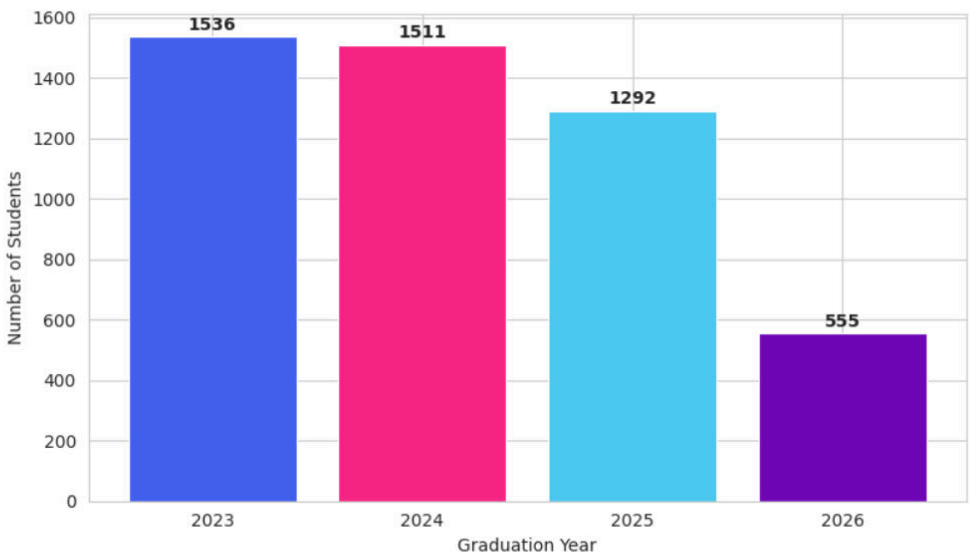
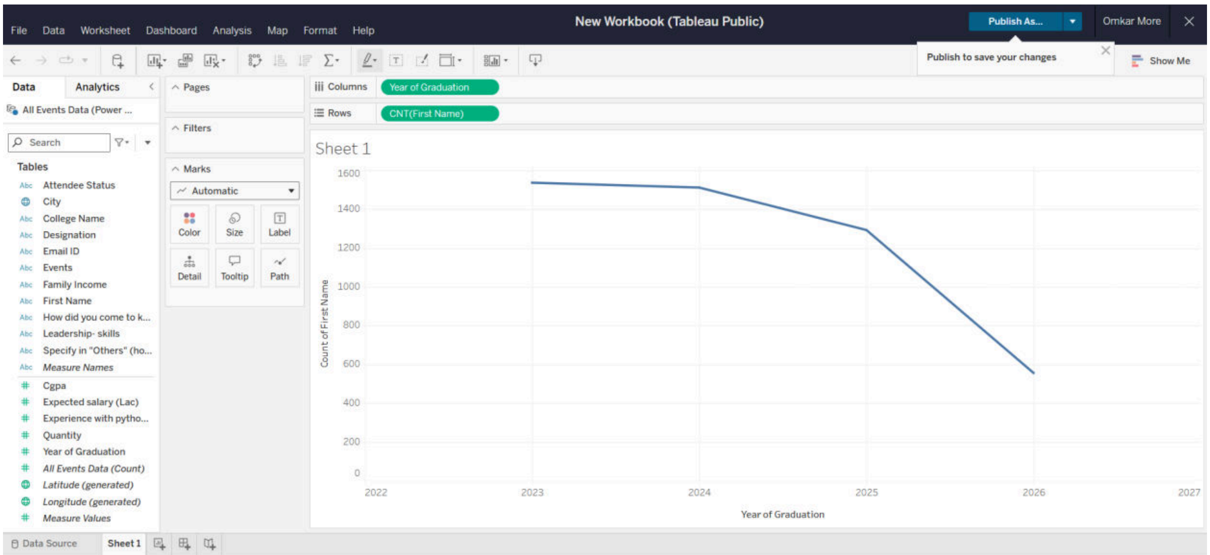


Tableau Output :



4. What is the distribution of student's experience with Python programming?

CODE :

```
py_dist = df['Experience with python (Months)'].value_counts().sort_index()
print(f"\n Python Experience Distribution (months):\n{py_dist.to_string()}")
fig, ax = plt.subplots(figsize=(8, 5))
```

ax.bar(py_dist.index.astype(str), py_dist.values, color='#4361ee', edgecolor='white')

ax.set_title("Q4 — Python Experience Distribution (Months)", fontsize=14, fontweight='bold', pad=15)

ax.set_xlabel("Experience (Months)"); ax.set_ylabel("Number of Students")

plt.tight_layout()

plt.show()

CONCLUSION : (Colab output)

```
Python Experience Distribution (months):
```

```
Experience with python (Months)
3    1008
4     466
5    1242
6     738
7     640
8     800
```

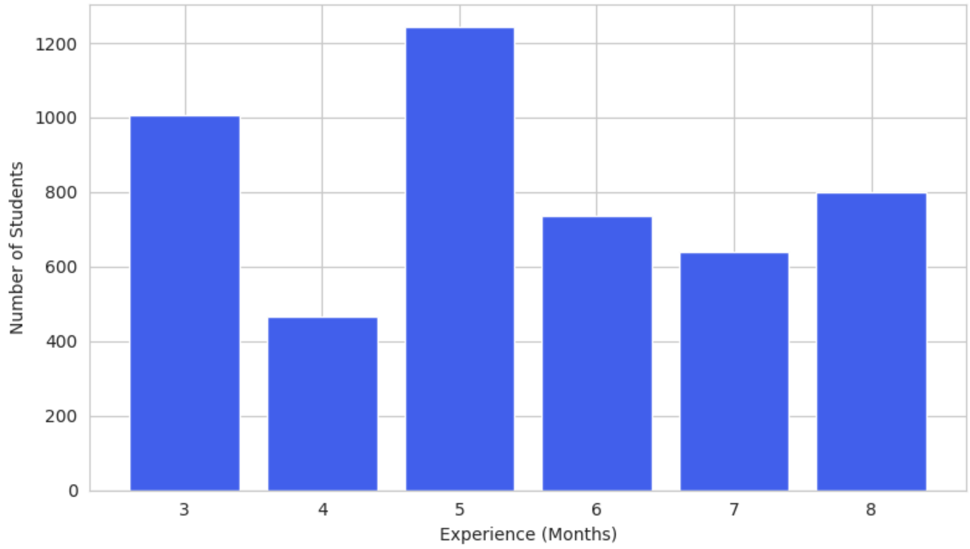
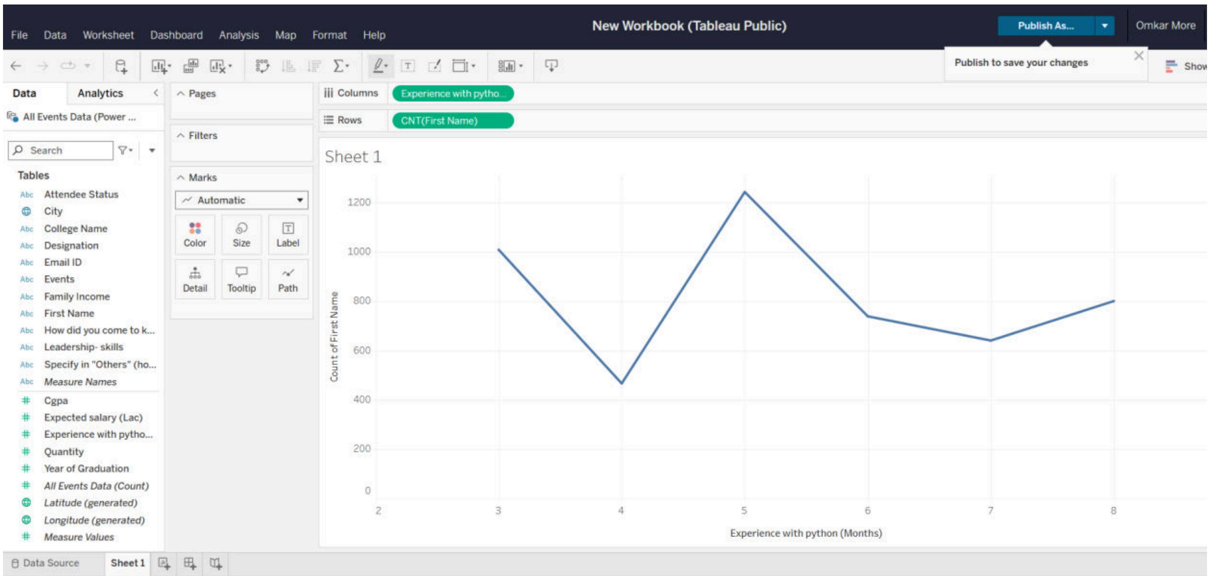


TABLEAU OUTPUT :



5. What is the average family income of the student?

CODE :

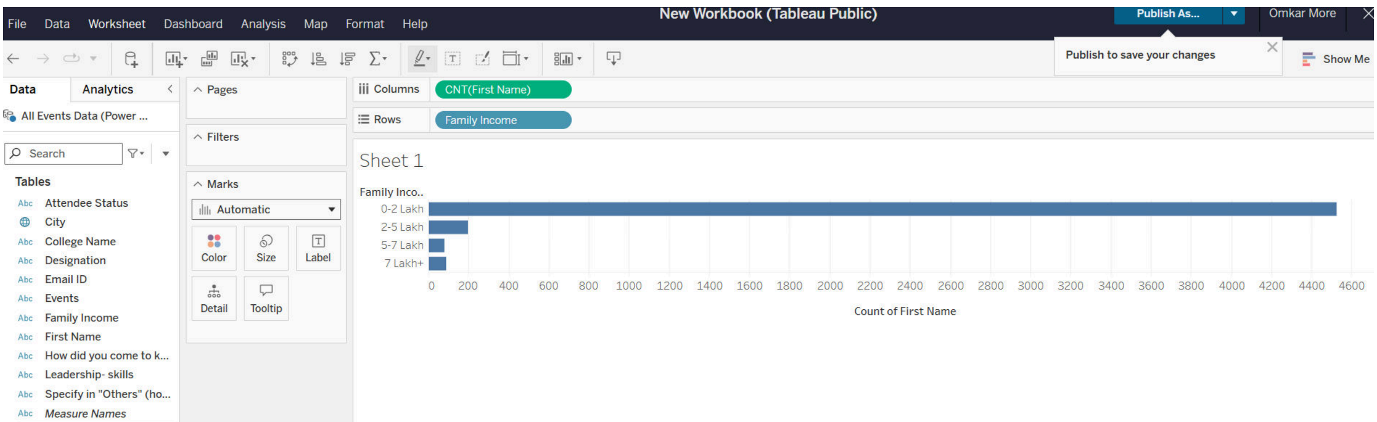
```
income_dist = df['Family Income'].value_counts()

avg_income_num = df['Family Income Numeric'].mean()

print(f"\n Family Income Distribution:\n{income_dist.to_string()}")

print(f"     Avg numeric income bracket midpoint: ₹{avg_income_num:.2f} Lakh")
```

CONCLUSION :

```
Family Income Distribution:
```

```
Family Income
0-2 Lakh     4525
2-5 Lakh      200
7 Lakh+        89
5-7 Lakh       80
       Avg numeric income bracket midpoint: ₹1.31 Lakh
```

TABLEAU OUTPUT :



6. How does the GPA vary among different colleges? *(Show top 5 results only)*

CODE :

top5_colleges = df.groupby('College Name')['CGPA'].mean().sort_values(ascending=False).head(5)

print(f"\nTop 5 Colleges by Avg GPA:\n{top5_colleges.to_string()}")

CONCLUSION : (Colab output)

```
Top 5 Colleges by Avg GPA:
```

```
College Name
THAKUR INSTITUTE OF MANAGEMENT STUDIES, CAREER DEVELOPMENT & RESEARCH - [TIMSCDR]    8.585714
St Xavier's College                                                                  8.578571
B. K. Birla College of Arts, Science & Commerce (Autonomous), Kalyan                 8.456410
Symbiosis Institute of Technology, Pune                                              8.303448
AP SHAH INSTITUTE OF TECHNOLOGY                                                      8.283333
```

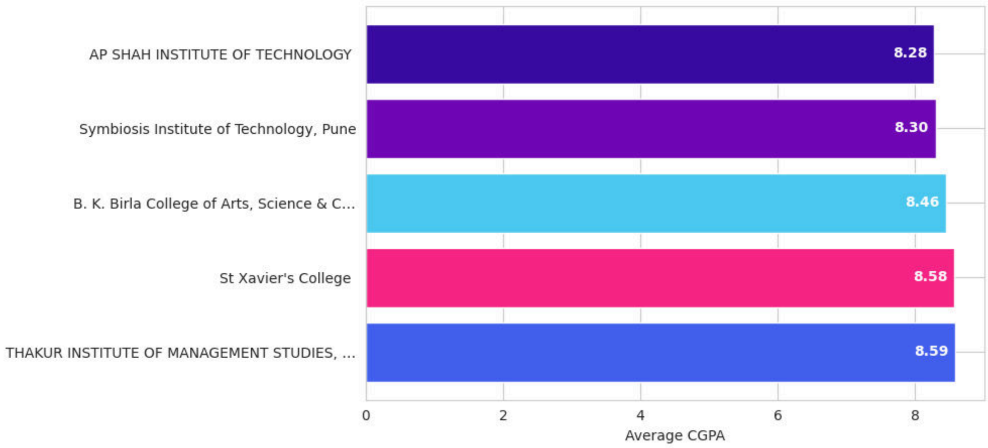
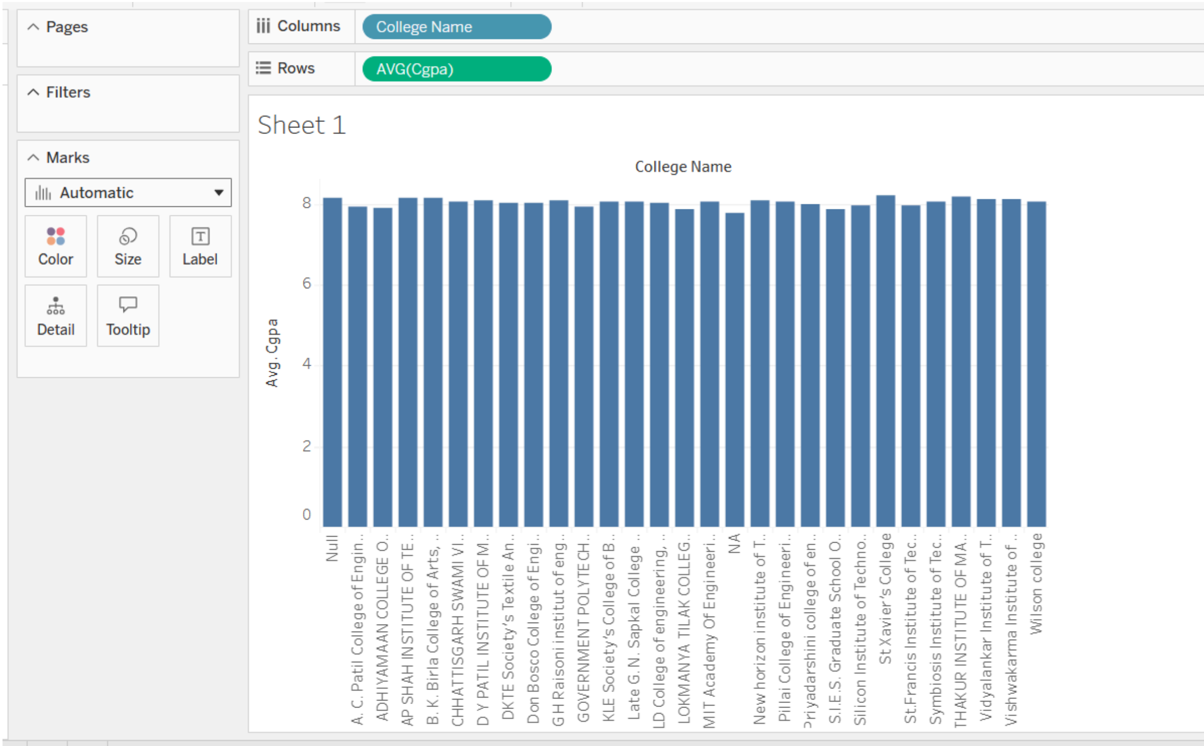


Tableau Output :



7. Are there any outliers in the quantity (number of courses completed) attribute?

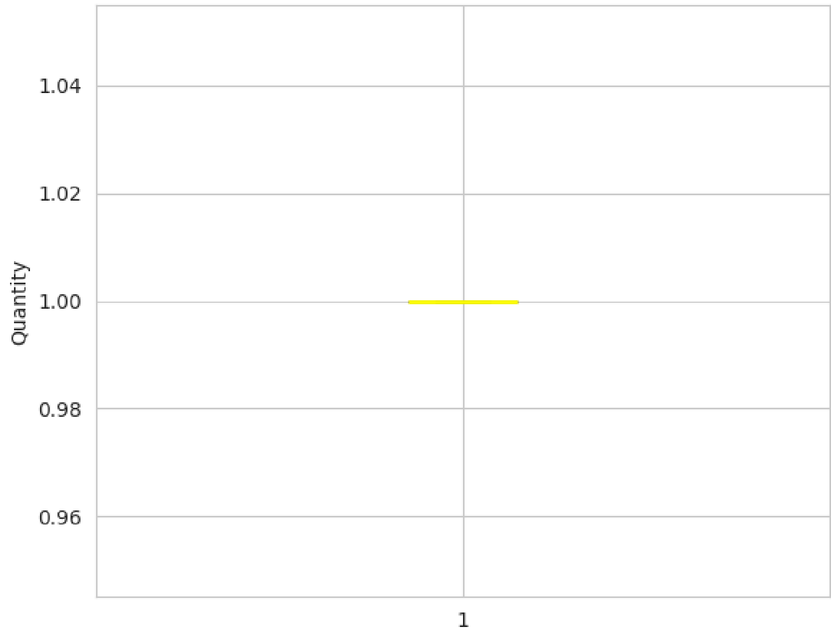
CODE :

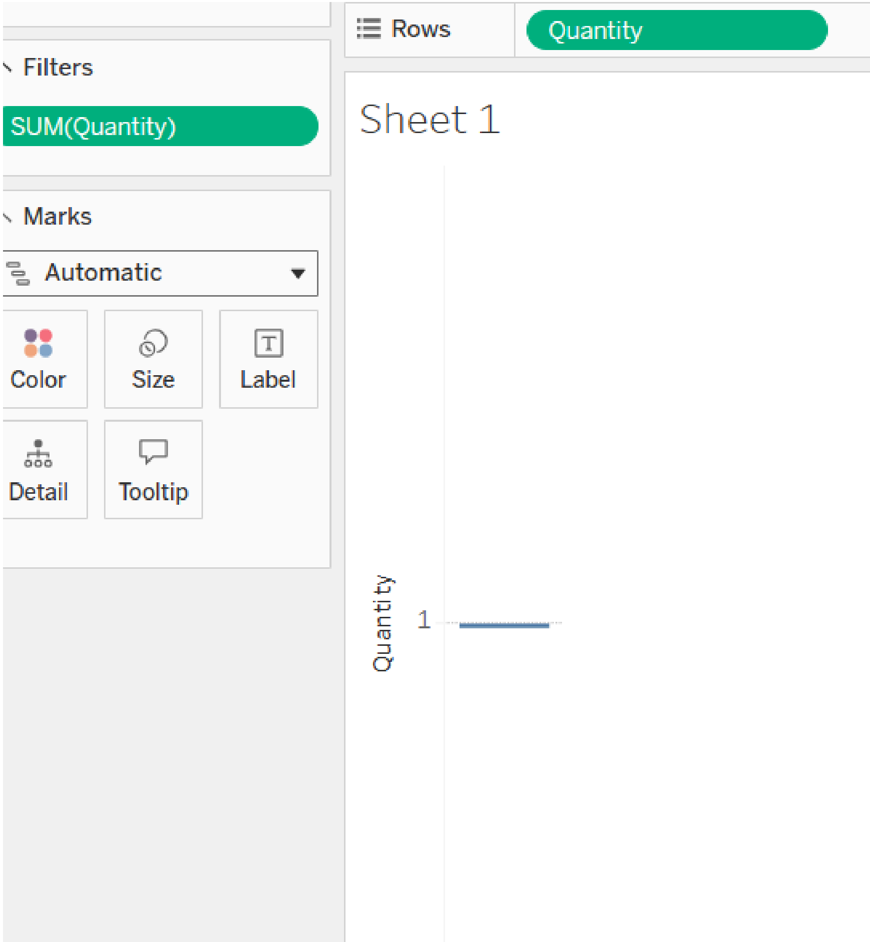
```
Q1 = df['Quantity'].quantile(0.25)

Q3 = df['Quantity'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR

upper = Q3 + 1.5 * IQR

outliers = df[(df['Quantity'] < lower) | (df['Quantity'] > upper)]

print(f"\n  Outlier Detection in Quantity:")

print(f"     IQR = {IQR}, Lower bound = {lower}, Upper bound = {upper}")

print(f"     Number of outliers: {len(outliers)}")

print(f"     Outlier values: {outliers['Quantity'].unique()}")

fig, ax = plt.subplots(figsize=(6, 5))

ax.boxplot(df['Quantity'], patch_artist=True,

        boxprops=dict(facecolor='#4361ee', color='navy'),

        medianprops=dict(color='yellow', linewidth=2),

        flierprops=dict(marker='o', markerfacecolor='#f72585', markersize=6))
```

```
ax.set_title("Q7 — Boxplot: Quantity (Courses Completed)", fontsize=13, fontweight='bold', pad=15)

ax.set_ylabel("Quantity")

plt.tight_layout()

plt.show()
```

CONCLUSION :

```
Outlier Detection in Quantity:
IQR = 0.0, Lower bound = 1.0, Upper bound = 1.0
Number of outliers: 0
Outlier values: []
```





8. What is the average GPA for student from each city?

CODE :

```
city_gpa = df.groupby('City')['CGPA'].mean().sort_values(ascending=False).head(10)

print(f"\n Average GPA by City (Top 10):\n{city_gpa.to_string()}")

fig, ax = plt.subplots(figsize=(10, 5))

bars = ax.bar(city_gpa.index, city_gpa.values, color=COLORS, edgecolor='white')

for bar in bars:

    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() - 0.3,

            f'{bar.get_height():.2f}', ha='center', va='top', color='white', fontsize=9,
fontweight='bold')

ax.set_title("Q8 — Avg GPA per City (Top 10)", fontsize=14, fontweight='bold', pad=15)

ax.set_xlabel("City"); ax.set_ylabel("Average CGPA")

plt.xticks(rotation=30, ha='right')

plt.tight_layout()

plt.show()
```

CONCLUSION : Colab Output :

```
Average GPA by City (Top 10):
```

```
City
Kolhapur     8.557143
Raipur       8.507143
Sonipat      8.464286
Gurugram     8.459259
Puri         8.450000
Siwan        8.450000
Srinagar     8.435714
Delhi        8.414286
Pune         8.400000
Hasan        8.392857
```

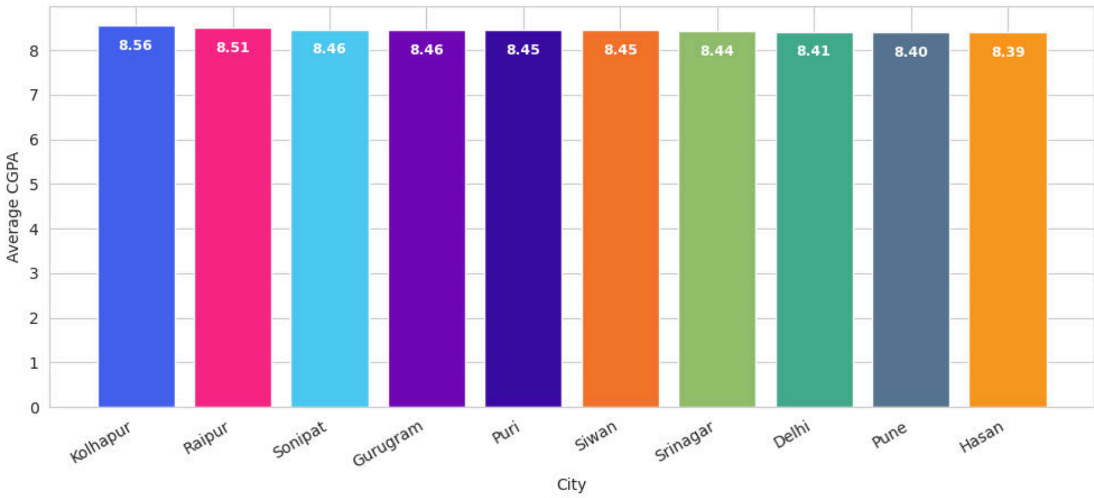
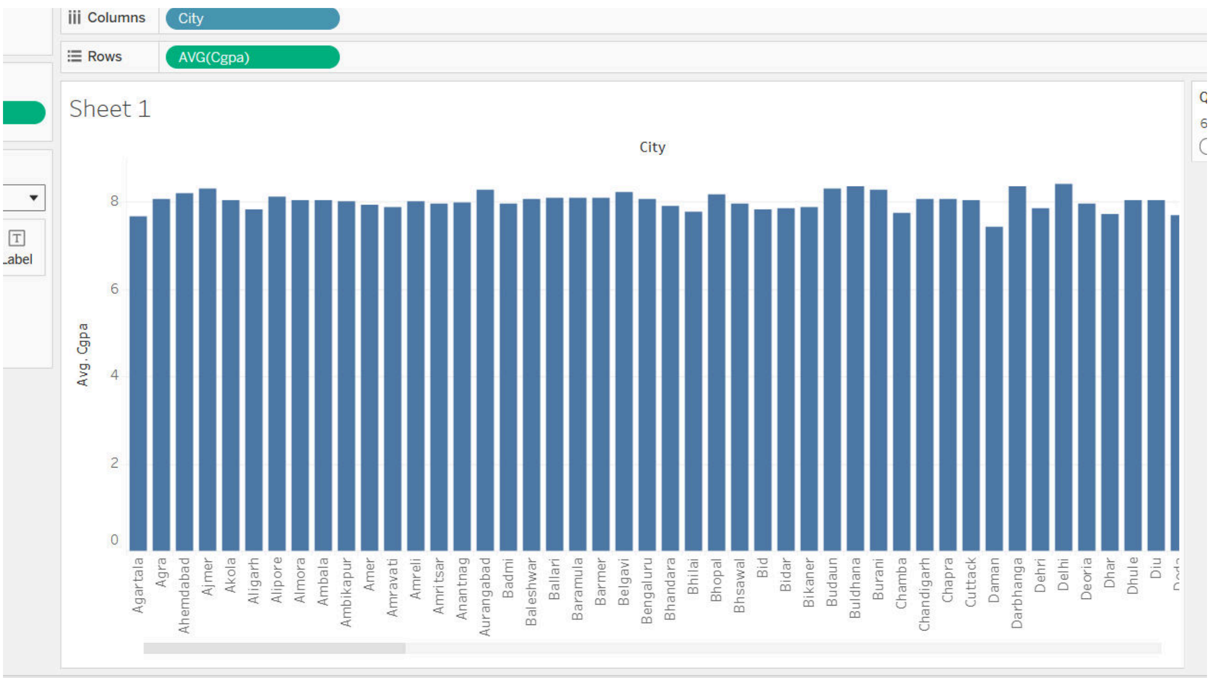


Tableau Output :



9. Can we identify any relationship between family income and GPA?

CODE :

```
corr = df[['Family Income Numeric', 'CGPA']].corr().iloc[0, 1]

print(f"\n Correlation between Family Income & GPA: {corr:.4f}")

print(f"      Interpretation: {'Weak' if abs(corr) < 0.3 else 'Moderate' if abs(corr) < 0.6 else 'Strong'} {'positive' if corr > 0 else 'negative'} correlation")


fig, ax = plt.subplots(figsize=(8, 5))

income_gpa = df.groupby('Family Income')['CGPA'].mean().reindex(['0-2 Lakh', '2-5 Lakh', '5-7 Lakh', '7 Lakh+'])

bars = ax.bar(income_gpa.index, income_gpa.values, color=COLORS[:4], edgecolor='white')

for bar in bars:

    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() - 0.3,

            f'{bar.get_height():.2f}', ha='center', va='top', color='white', fontweight='bold')

ax.set_title(f"Q9 — Avg GPA by Family Income  (Corr: {corr:.3f})", fontsize=13, fontweight='bold', pad=15)

ax.set_xlabel("Family Income"); ax.set_ylabel("Average CGPA")

plt.tight_layout()

plt.show()

print("\n" + "=" * 60)

print("  MODERATE QUESTIONS")

print("=" * 60)
```

CONCLUSION : Colab Output :

```
 Correlation between Family Income & GPA: 0.0157
Interpretation: Weak positive correlation
```

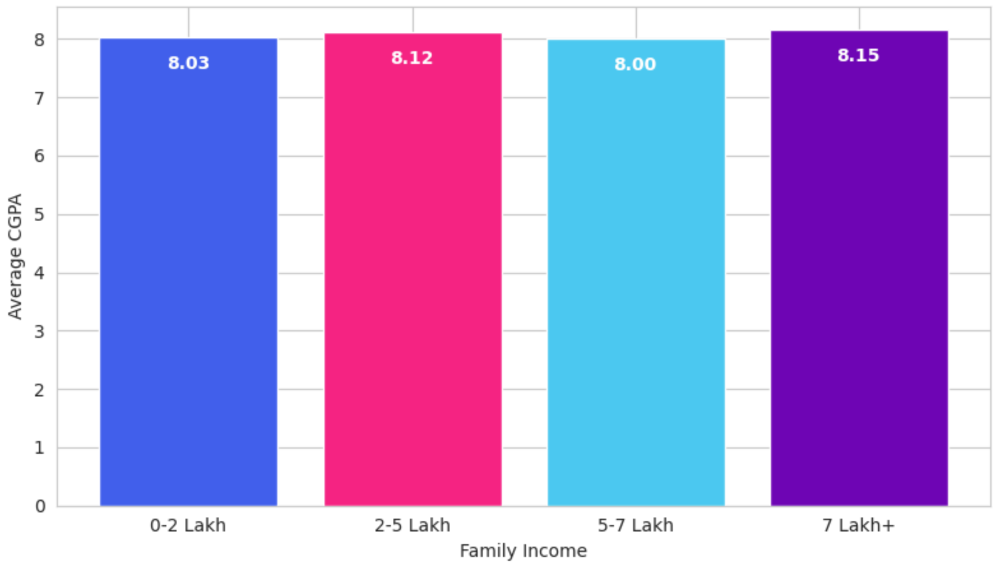
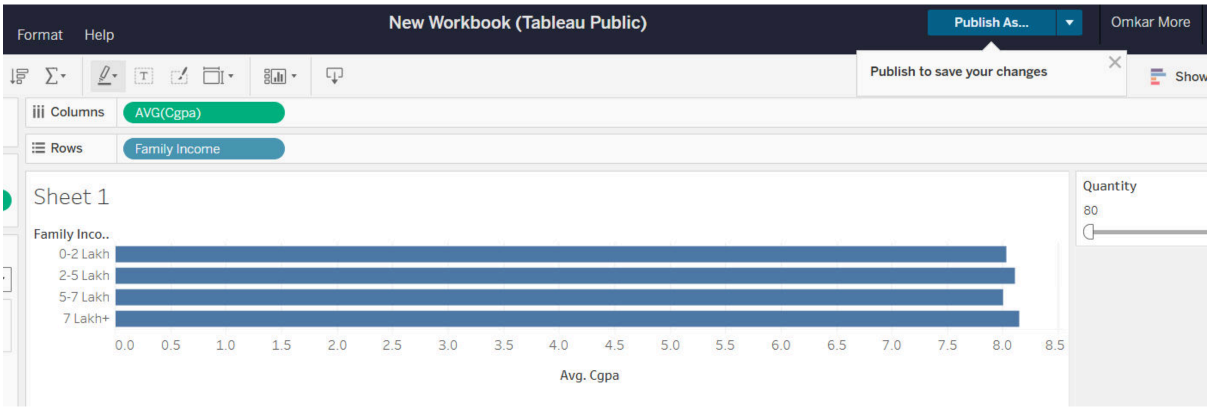


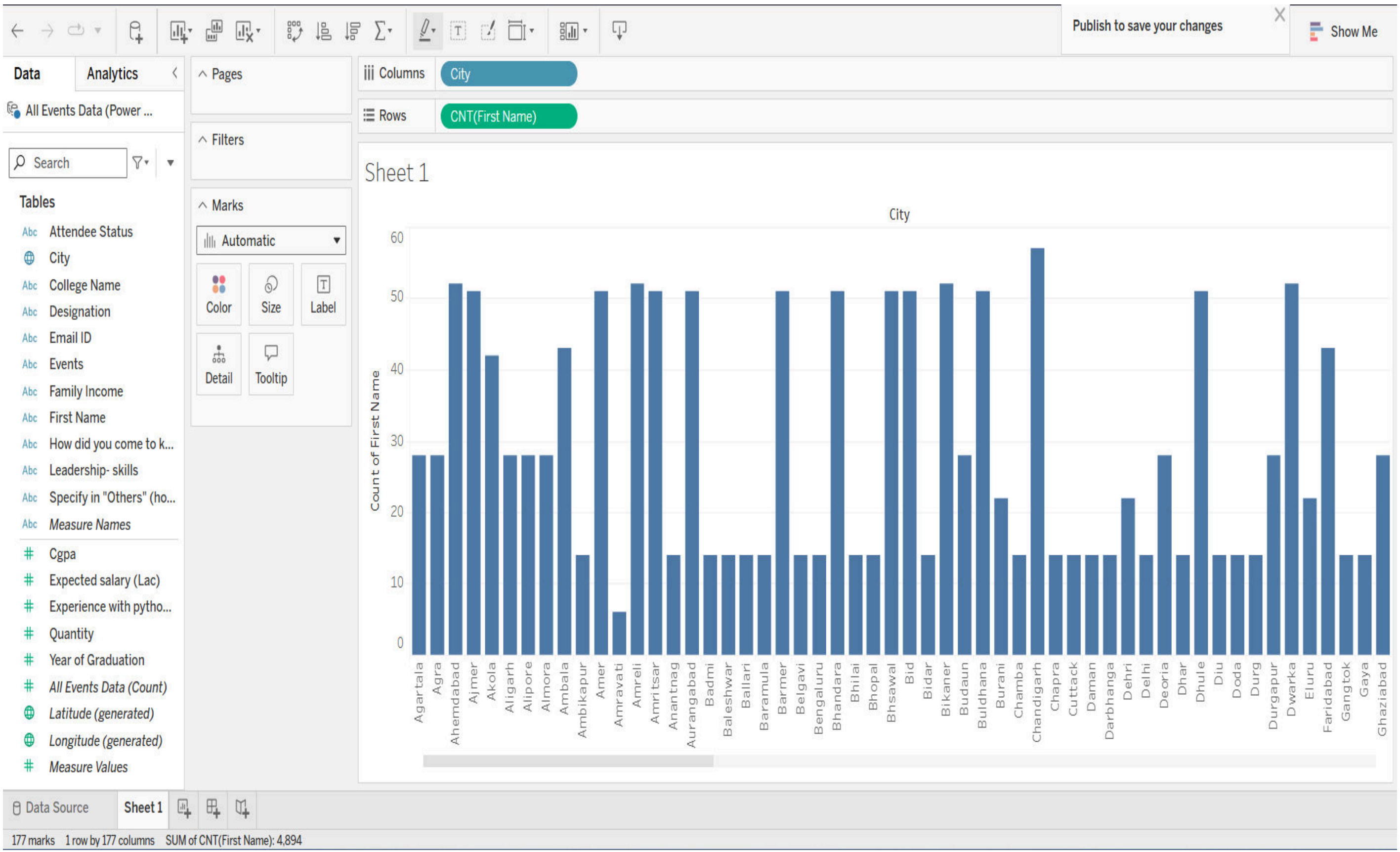
Tableau Output :



(B) MODERATE QUESTIONS (Attempt any 8):

10. How many students from various cities? *(Solve using data visualisation tool)*

CONCLUSION : (BY USING TABLEU PUBLIC TOOL USED FOR DATA VISUALIZATION)



CODE : (on Google Colab only top 15 cities)

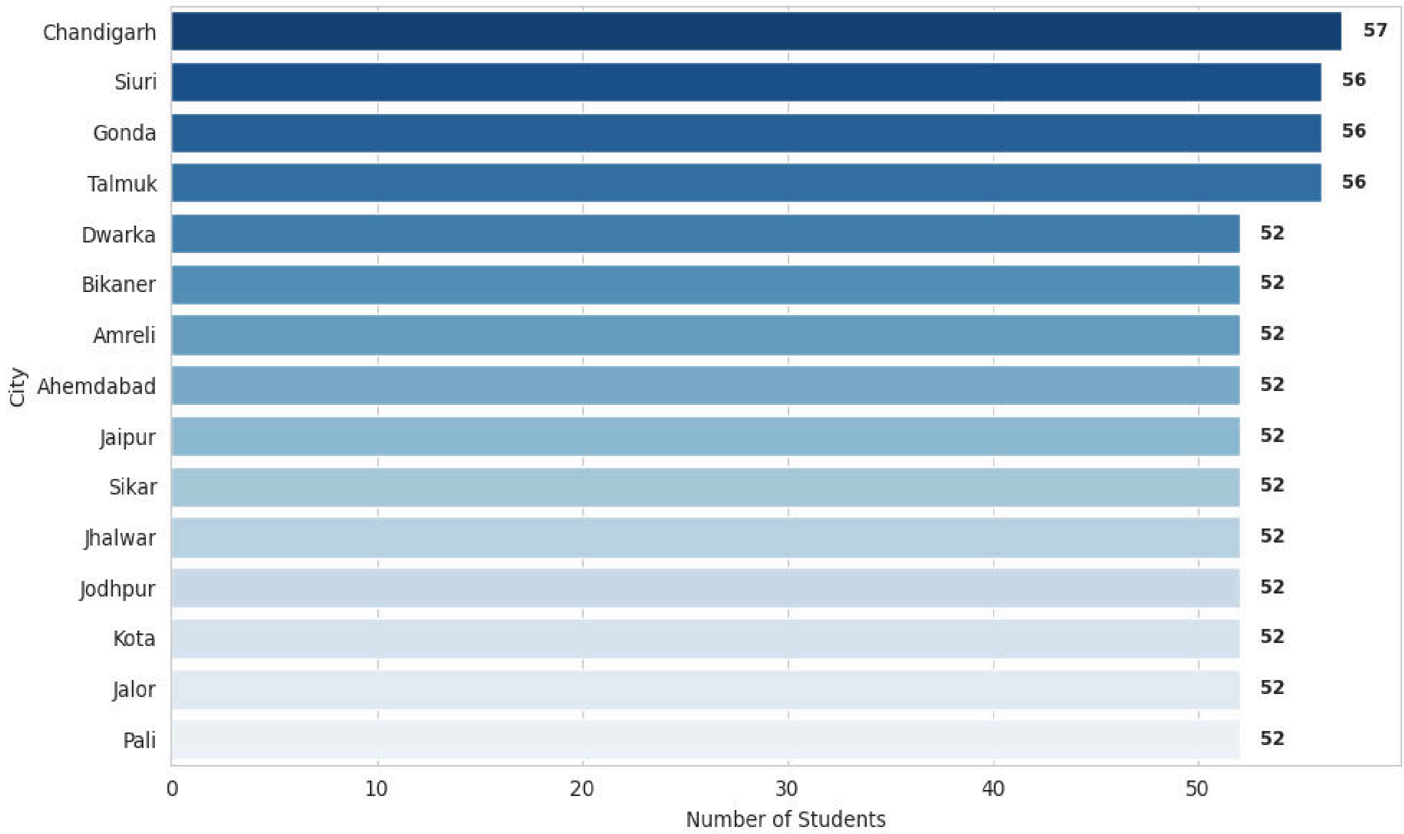
```
city_counts = df['City'].value_counts().head(15)

print(f"\n Top 15 Cities by Student Count:\n{city_counts.to_string()}")

fig, ax = plt.subplots(figsize=(10, 6))

sns.barplot(x=city_counts.values, y=city_counts.index, palette='Blues_r', ax=ax)

for i, v in enumerate(city_counts.values):

    ax.text(v + 1, i, str(v), va='center', fontsize=9, fontweight='bold')

ax.set_title("Q10 — Students from Various Cities (Top 15)", fontsize=14, fontweight='bold',
pad=15)

ax.set_xlabel("Number of Students"); ax.set_ylabel("City")

plt.tight_layout()
```

plt.show()

CONCLUSION : (ON GOOGLE COLAB) :

```
Top 15 Cities by Student Count:
```

```
City
Chandigarh    57
Siuri         56
Gonda         56
Talmuk        56
Dwarka        52
Bikaner       52
Amreli        52
Ahemdabad     52
Jaipur        52
Sikar         52
Jhalwar       52
Jodhpur       52
Kota          52
Jalor         52
Pali          52
```



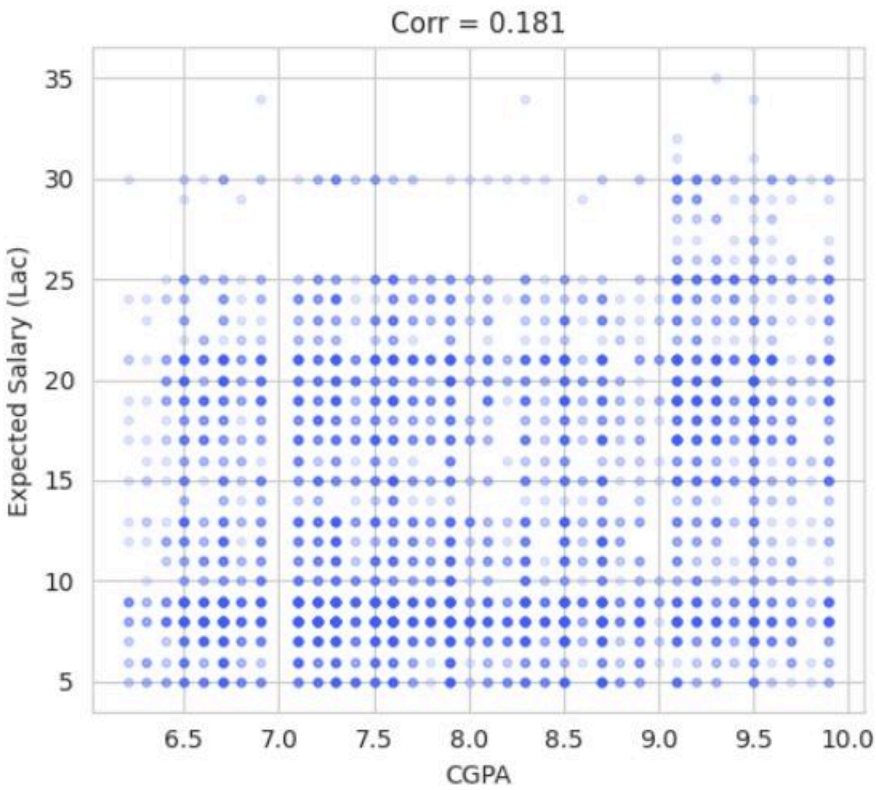
11. How does the expected salary vary based on factors like 'GPA', 'Family income', 'Experience with Python (Months)'?

CODE :

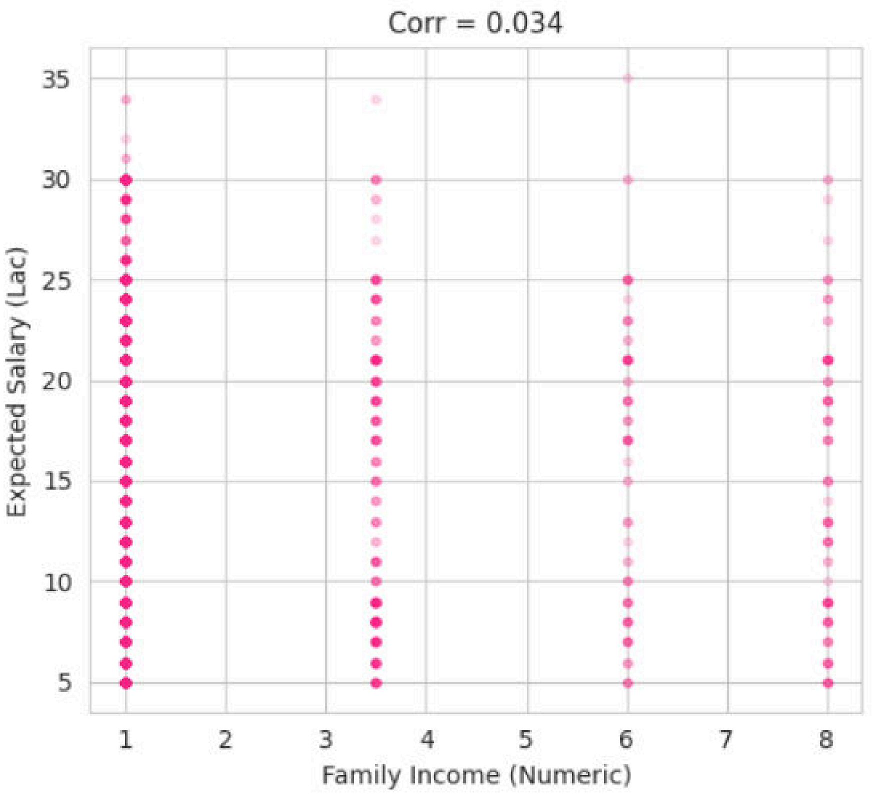
```
print(f"\n  Expected Salary vs Key Factors:")
corr_gpa_sal   = df[['CGPA', 'Expected salary (Lac)']].corr().iloc[0, 1]
corr_inc_sal   = df[['Family Income Numeric', 'Expected salary (Lac)']].corr().iloc[0, 1]
corr_py_sal    = df[['Experience with python (Months)', 'Expected salary (Lac)']].corr().iloc[0, 1]
print(f"      GPA vs Expected Salary correlation:           {corr_gpa_sal:.4f}")
print(f"      Family Income vs Expected Salary correlation:  {corr_inc_sal:.4f}")
print(f"      Python Exp vs Expected Salary correlation:     {corr_py_sal:.4f}")
fig, axes = plt.subplots(1, 3, figsize=(15, 5))
for ax, col, label, corr_val, color in zip(
    axes,
    ['CGPA', 'Family Income Numeric', 'Experience with python (Months)'],
    ['CGPA', 'Family Income (Numeric)', 'Python Experience (Months)'],
    [corr_gpa_sal, corr_inc_sal, corr_py_sal],
    ['#4361ee', '#f72585', '#4cc9f0']
):
    ax.scatter(df[col], df['Expected salary (Lac)'], alpha=0.15, color=color, s=10)
    ax.set_xlabel(label); ax.set_ylabel("Expected Salary (Lac)")
    ax.set_title(f"Corr = {corr_val:.3f}", fontsize=11)
fig.suptitle("Q11 — Expected Salary vs Key Factors", fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```

CONCLUSION :

```
) Expected Salary vs Key Factors:
GPA vs Expected Salary correlation:           0.1812
Family Income vs Expected Salary correlation:  0.0344
Python Exp vs Expected Salary correlation:     0.1156
```



Corr = 0.181
Expected Salary (Lac)
CGPA
35
30
25
20
15
10
5
6.5
7.0
7.5
8.0
8.5
9.0
9.5
10.0



Corr = 0.034
Expected Salary (Lac)
Family Income (Numeric)
35
30
25
20
15
10
5
1
2
3
4
5
6
7
8

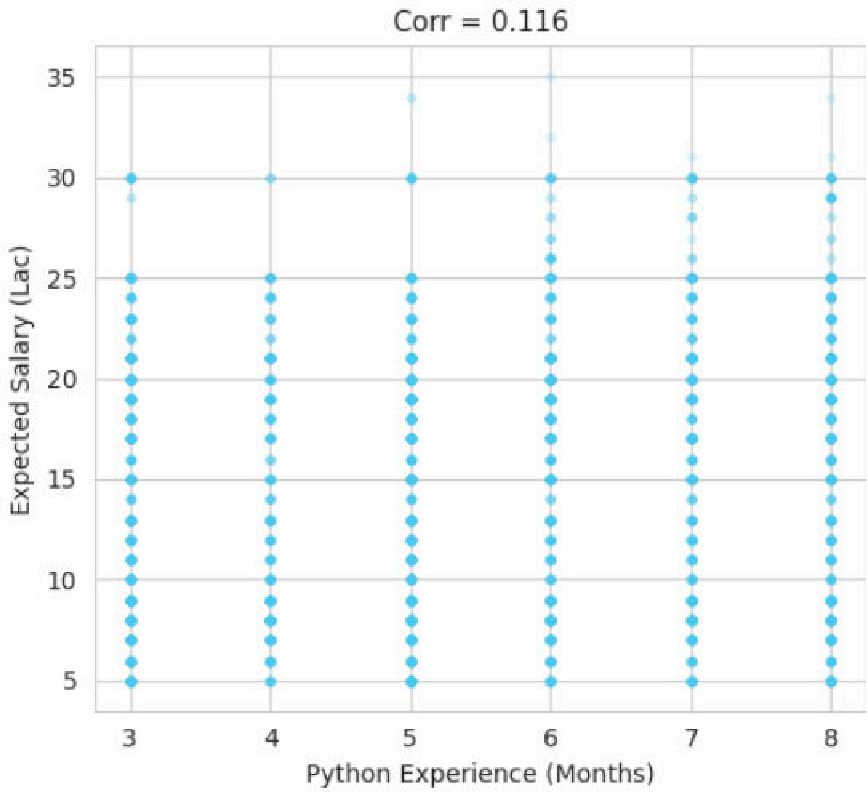
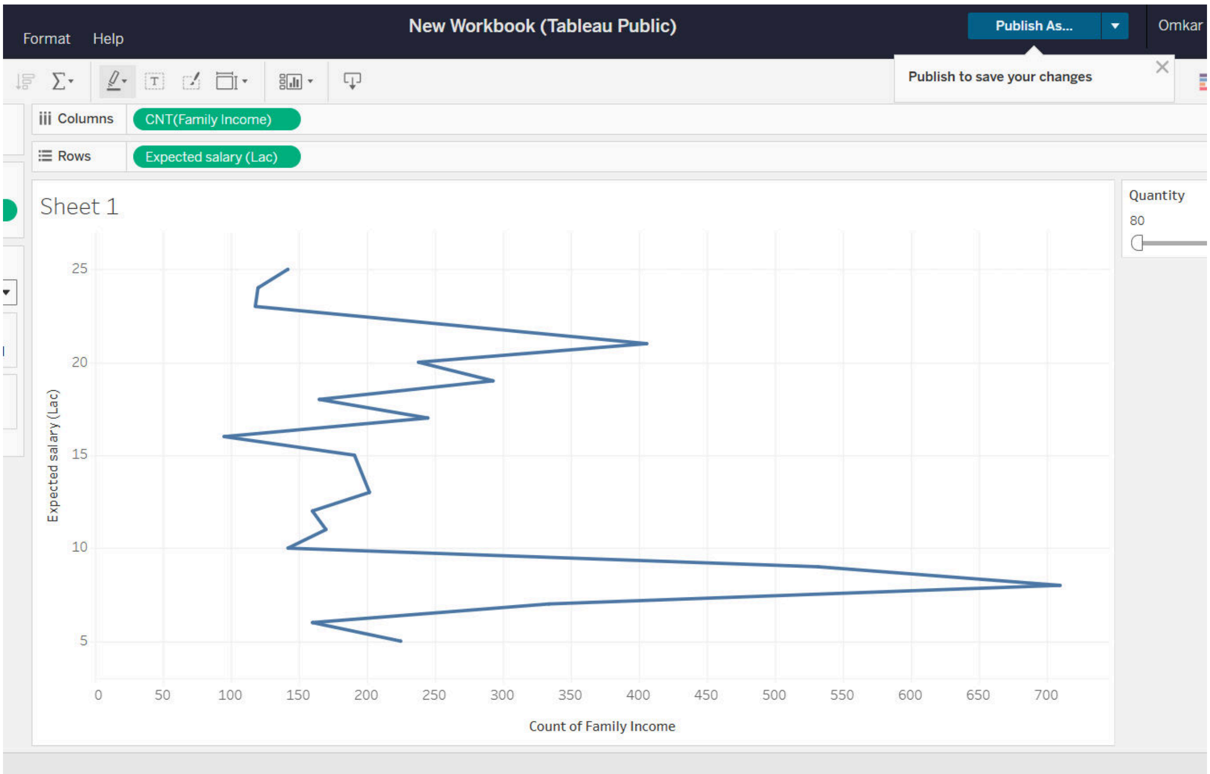
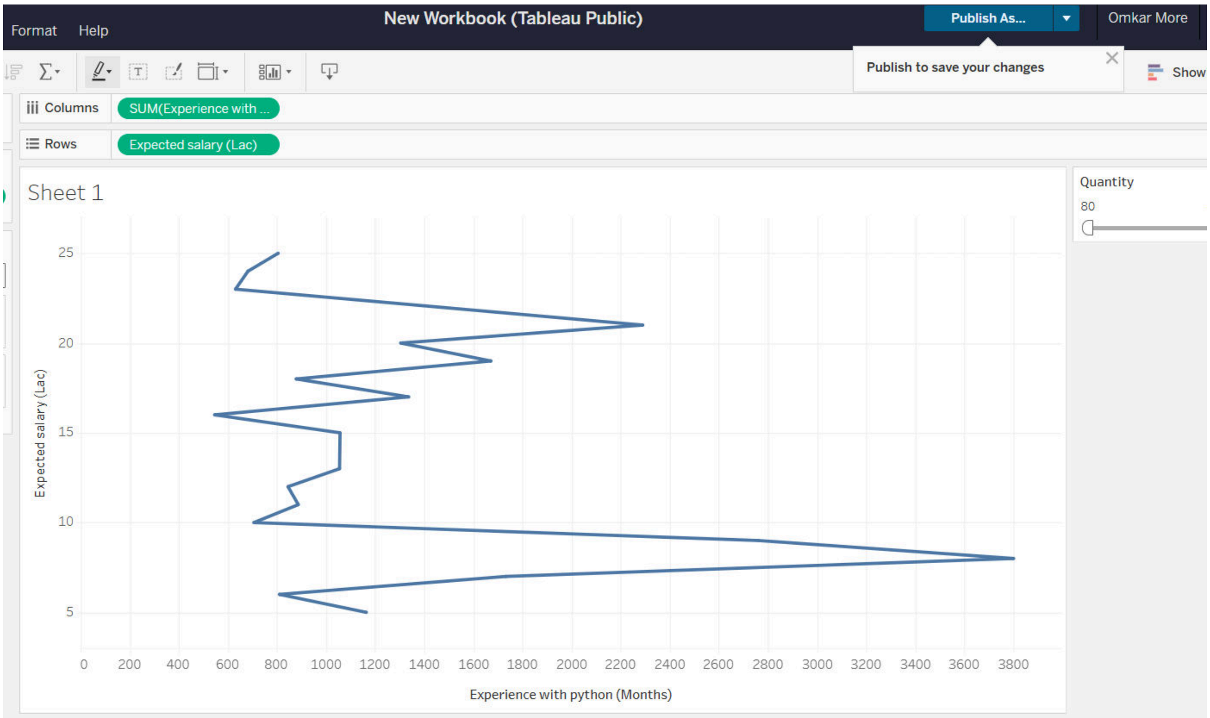


Tableau Output :





12. Which event tend to attract more students from specific fields of study?

CODE :

```
event_college = df.groupby('Events')['College Name'].nunique().sort_values(ascending=False)
```

print(f"\n Events by number of unique colleges:\n{event_college.to_string()}")

fig, ax = plt.subplots(figsize=(10, 6))

sns.barplot(x=event_college.values, y=event_college.index, palette='viridis', ax=ax)

ax.set_title("Q12 — Events vs Unique Colleges Attracted", fontsize=13, fontweight='bold', pad=15)

ax.set_xlabel("Unique Colleges"); ax.set_ylabel("Event")

plt.tight_layout()

plt.show()

CONCLUSION : Colab Output :

```
Events by number of unique colleges:
Events
Art of Resume Building                           55
Internship Program(IP) Success Conclave          54
Data Visualization using Power BI                53
Artificial Intelligence                          52
IAC - Q&A                                        35
Hello ML and DL                                  27
IS DATA SCIENCE FOR YOU?                         27
KYC - Know Your CCPC                             27
Product Design & Full Stack                      27
RPA: A Boon or A Bane                            27
Skill and Employability Enhancement              27
Talk on Skill and Employability Enhancement      27
The SDLC & their transformations                 27
The Agile Ways of Working                        27
Product Marketing                                24
Transformation with DevOps: The Easy Way         24
```

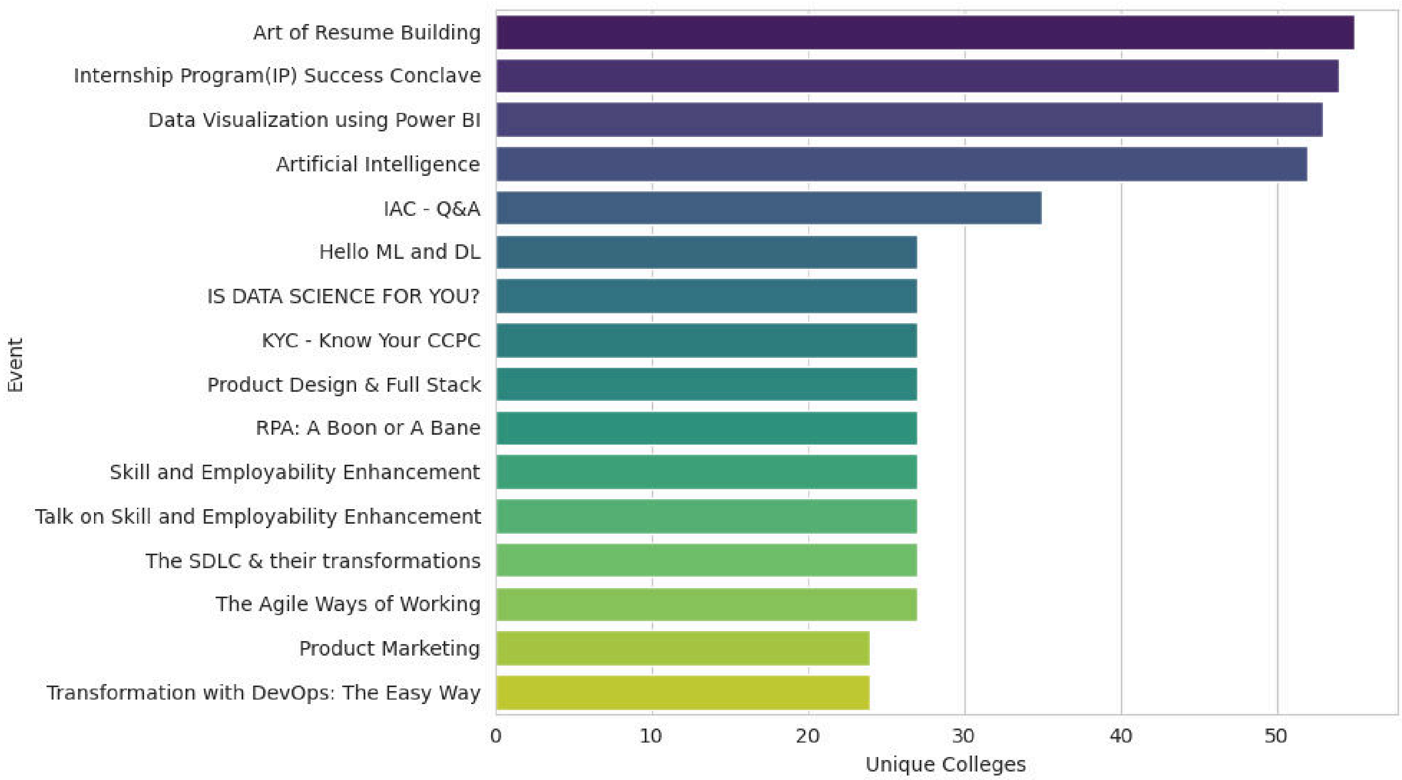


Tableau Output :

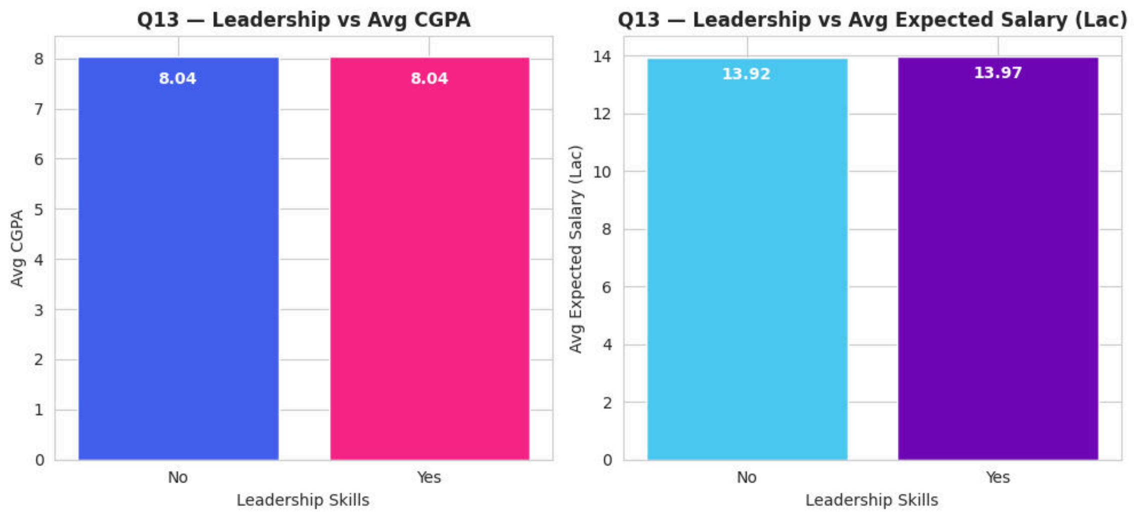
13. Do students in leadership positions during their college years tend to have higher GPAs or better expected salary?

CODE :

```
leadership = df[df['Leadership- skills'].isin(['yes', 'no'])]

leader_gpa = leadership.groupby('Leadership- skills')['CGPA'].mean()

leader_sal = leadership.groupby('Leadership- skills')['Expected salary (Lac)'].mean()

print(f"\n  Leadership Skills vs GPA & Salary:")

print(f"      Avg GPA — Yes: {leader_gpa.get('yes', 0):.2f}  |  No: {leader_gpa.get('no', 0):.2f}")

print(f"      Avg Salary — Yes: {leader_sal.get('yes', 0):.2f}  |  No: {leader_sal.get('no', 0):.2f}")

fig, axes = plt.subplots(1, 2, figsize=(10, 5))

for ax, data, title, color_pair in zip(

    axes,

    [leader_gpa, leader_sal],

    ['Avg CGPA', 'Avg Expected Salary (Lac)'],

    [['#4361ee', '#f72585'], ['#4cc9f0', '#7209b7']]

):

    bars = ax.bar(data.index.str.title(), data.values, color=color_pair, edgecolor='white')

    for bar in bars:
```

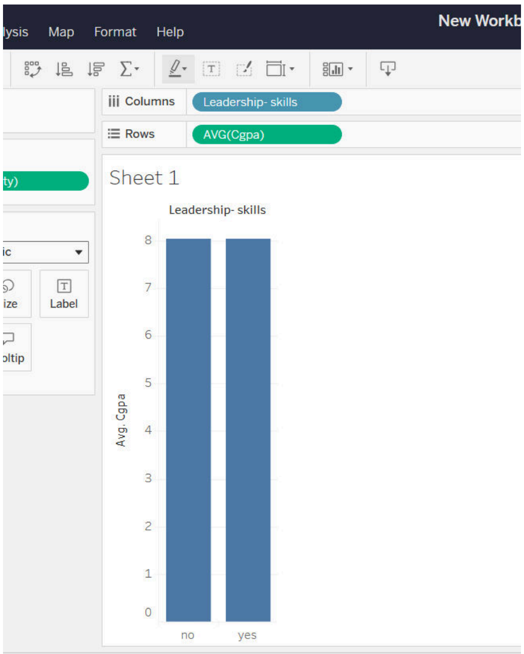
```
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() - 0.3,
                f'{bar.get_height():.2f}', ha='center', va='top', color='white', fontweight='bold')
    ax.set_title(f"Q13 — Leadership vs {title}", fontsize=12, fontweight='bold')
    ax.set_xlabel("Leadership Skills"); ax.set_ylabel(title)
fig.suptitle("Q13 — Leadership Skills Impact", fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

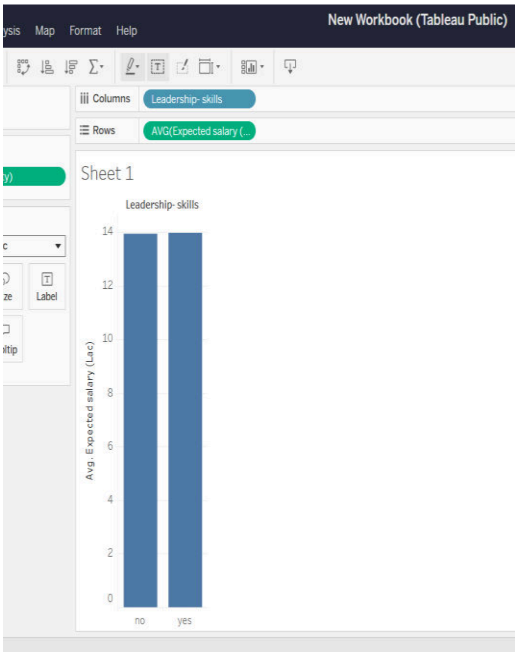
CONCLUSION : Colab Output :




```
) Leadership Skills vs GPA & Salary:
Avg GPA — Yes: 8.04  |  No: 8.04
Avg Salary — Yes: 13.97  |  No: 13.92
```

Tableau Output :





14. How many students are graduating by the end of 2024?

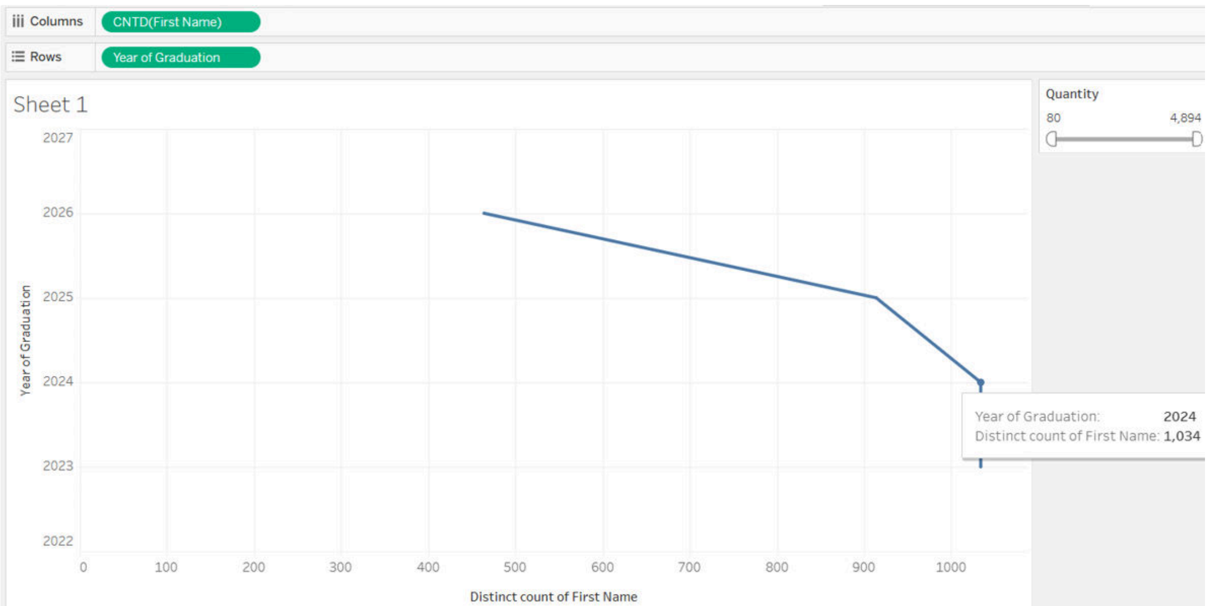
CODE :

```
graduating_2024 = df[df['Year of Graduation'] <= 2024]['Email ID'].nunique()

print(f"\n  Unique students graduating by end of 2024: {graduating_2024}")
```

CONCLUSION :

```
Unique students graduating by end of 2024: 1664
```

Tableau Output :



15. Which promotion channel brings in more student participations for the event?

CODE :

CONCLUSION :

```
Top 10 Promotion Channels:
How did you come to know about this event?
Whatsapp                              1067
Email                                  438
SPOC/ College Professor                326
Others                                 153
Cloud Counselage Website               129
Whatsapp | SPOC/ College Professor      67
LinkedIn                                55
Facebook                                48
Youtube                                 37
Friend/ Classmate                       30
```

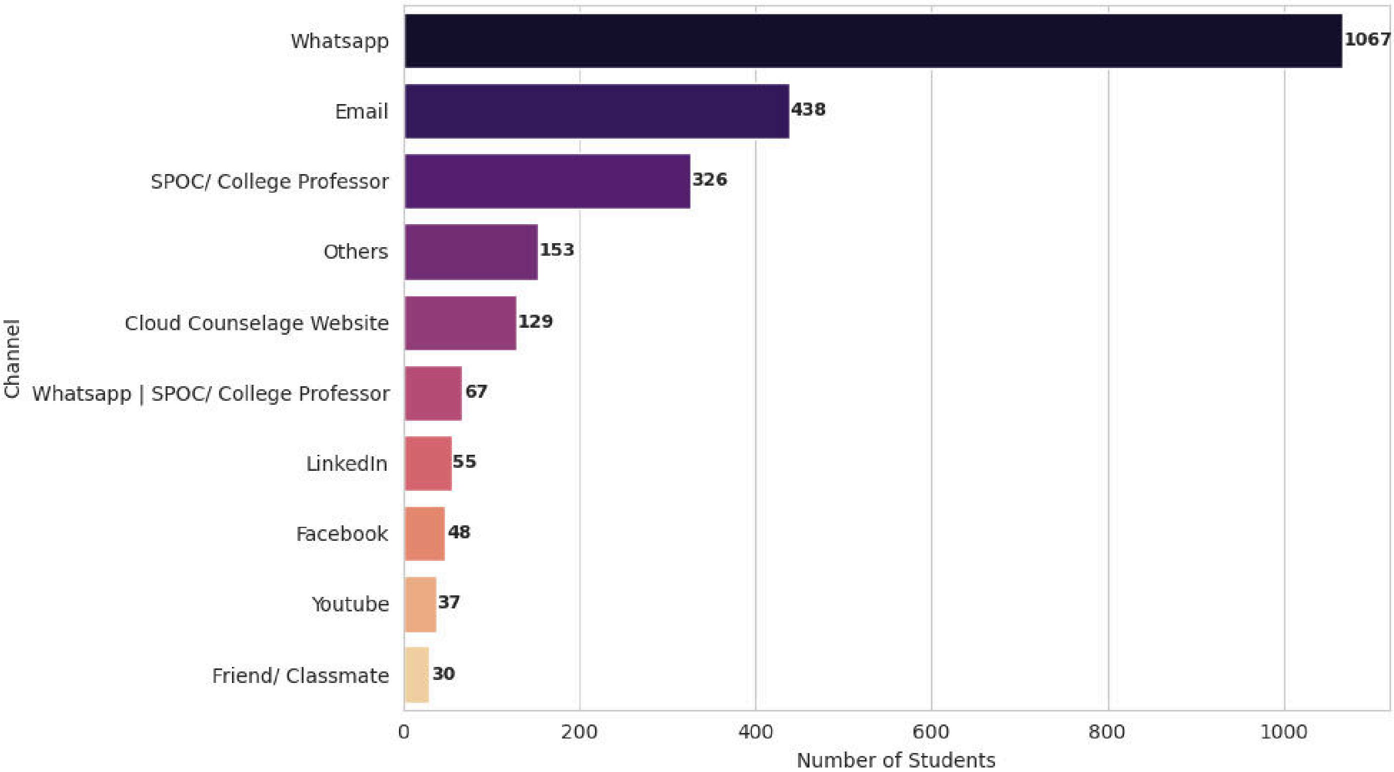
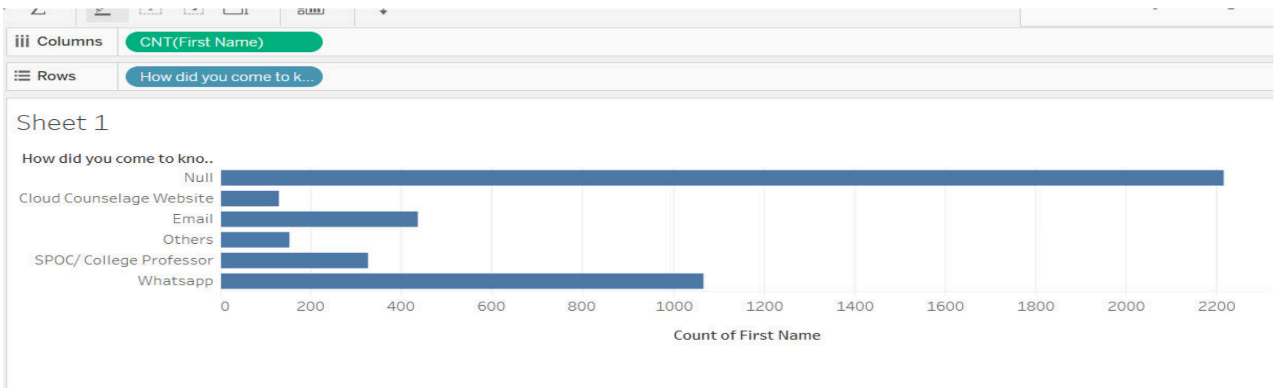


Tableau Output :



16. Find the total number of students who attended the events related to Data Science? *(From all Data Science related courses)*

CODE :

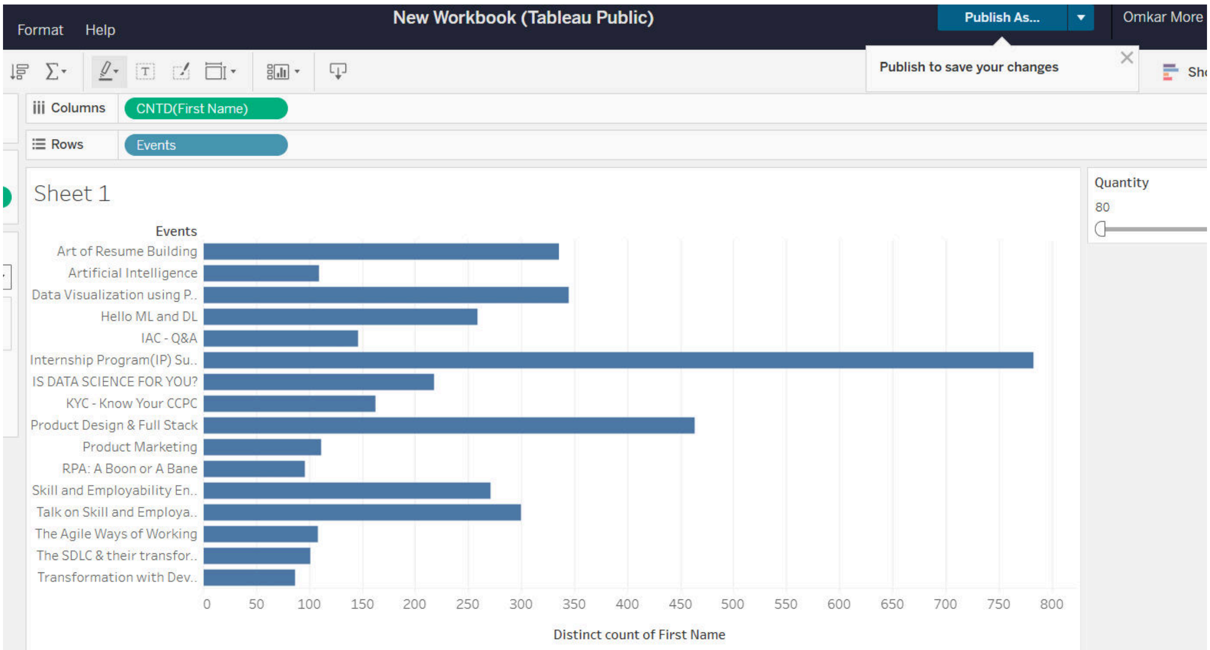
```
ds_keywords = ['data science', 'data visualization', 'hello ml', 'artificial intelligence']

ds_mask = df['Events'].str.lower().apply(lambda e: any(kw in e for kw in ds_keywords))

ds_students = df[ds_mask]['Email ID'].nunique()

ds_event_counts = df[ds_mask]['Events'].value_counts()

print(f"\n  Total unique students in Data Science related events: {ds_students}")

print(f"      Events counted:\n{ds_event_counts.to_string()}")
```

CONCLUSION : Colab Output :

```
Total unique students in Data Science related events: 826
```

```
       Events counted:
Events
Data Visualization using Power BI    455
IS DATA SCIENCE FOR YOU?             306
Hello ML and DL                      262
Artificial Intelligence              125
```

Tableau Output :



17. Those who have high CGPA & more experience in language — those who had high expectations for salary? *(Avg)*

CODE :

```
cgpa_median = df['CGPA'].median()
py_median   = df['Experience with python (Months)'].median()
high_performers = df[(df['CGPA'] >= cgpa_median) & (df['Experience with python
(Months)']>= py_median)]
avg_salary_hp = high_performers['Expected salary (Lac)'].mean()
overall_avg   = df['Expected salary (Lac)'].mean()
print(f"\n High CGPA (≥{cgpa_median}) & High Python Exp (≥{py_median} months):")
print(f"      Count: {len(high_performers)}")
print(f"      Avg Expected Salary: ₹{avg_salary_hp:.2f} Lac")
print(f"      vs Overall Avg Salary: ₹{overall_avg:.2f} Lac")
print(f"      Difference: ₹{avg_salary_hp - overall_avg:.2f} Lac more")
fig, ax = plt.subplots(figsize=(6, 5))
categories = ['All Students', 'High CGPA +\nHigh Python Exp']
values = [overall_avg, avg_salary_hp]
bars = ax.bar(categories, values, color=['#4361ee', '#f72585'], edgecolor='white', width=0.4)
for bar in bars:
ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() - 0.3,
f'₹{bar.get_height():.2f}L', ha='center', va='top', color='white', fontsize=12, fontweight='bold')
ax.set_title("Q17 — Salary: High Performers vs All", fontsize=13, fontweight='bold', pad=15)
ax.set_ylabel("Avg Expected Salary (Lac)")
plt.tight_layout()
plt.show()
print("\n" + "=" * 60)
print("=" * 60)
```

CONCLUSION :

```
) High CGPA (≥7.9) & High Python Exp (≥5.0 months):
Count: 1824
Avg Expected Salary: ₹15.14 Lac
vs Overall Avg Salary: ₹13.94 Lac
Difference: ₹1.20 Lac more
```

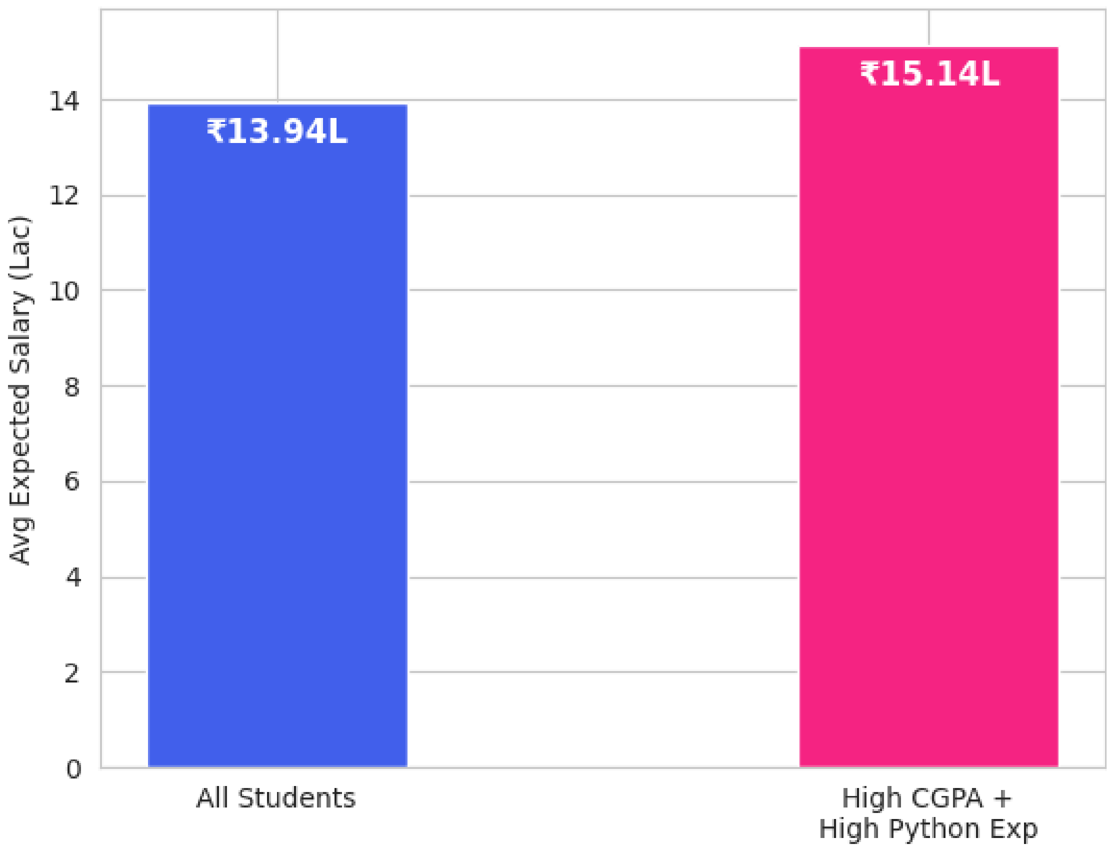


Tableau Output :